SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour
QUESTIONS: 4 **Total: Marks:40**

Annexure A

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Code of Conduct:

I Lokesh S (192511037) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Lokesh S

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

Inductive learning learns a general rule or predictive model from previously observed examples and applies that knowledge to new unseen cases. For this healthcare problem, a **supervised inductive learning approach** is suitable because the patient records contain input features along with a known diabetes outcome. A classification algorithm can learn the relationship between patient characteristics and diabetes diagnosis.

The **target variable** is **Diabetes Status**, which can be represented as 1 for diabetic and 0 for non-diabetic.

Important features include:

1. **Glucose Level** – High blood glucose is one of the strongest indicators of diabetes and is therefore an important predictor.
2. **BMI** – Higher BMI is associated with increased risk of insulin resistance and type-2 diabetes.
3. **Age** – Diabetes risk generally increases with age because metabolic and insulin-related changes become more common.
4. **Blood Pressure** – High blood pressure is associated with metabolic disorders and can provide useful information for diabetes risk prediction.
5. **Family History** – A family history of diabetes indicates genetic and hereditary risk.

Thus, the model learns patterns from these features and predicts whether a new patient is likely to have diabetes.

(b) Explain how you would handle class imbalance. Suggest and justify a suitable balancing strategy.

The dataset may contain significantly more non-diabetic patients than diabetic patients. This creates **class imbalance**, where a model may achieve high accuracy simply by predicting most patients as non-diabetic. However, this is dangerous in healthcare because diabetic patients may be missed.

I would use **SMOTE (Synthetic Minority Over-sampling Technique)** as the primary balancing strategy. SMOTE generates synthetic examples of the minority diabetic class by interpolating between existing minority-class samples.

For example, instead of simply duplicating diabetic records, SMOTE creates new realistic combinations of feature values around existing diabetic patients.

The procedure would be:

1. Separate the training and testing data.
2. Apply SMOTE **only to the training set**.
3. Generate additional diabetic samples.
4. Train the classifier using the balanced training data.
5. Evaluate the model on the original, untouched test set.

SMOTE is preferred because it helps the model learn the characteristics of the minority diabetic class without simply duplicating existing records. In a medical application, **Recall/Sensitivity** should also be monitored because missing a diabetic patient is more serious than generating an additional false alarm.

(c) Split the dataset (70:30) and justify the split ratio. State preprocessing steps and their impact.

The dataset can be divided into:

- **70% training data** – used to learn the patterns.
- **30% testing data** – used to evaluate performance on unseen patients.

A **70:30 split** provides enough data for training while retaining a sufficiently large independent test set for reliable evaluation. For medical data, the split should preferably be **stratified**, so that diabetic and non-diabetic cases maintain approximately the same proportion in both sets.

Preprocessing steps include:

1. Missing-value handling:

Missing numerical values can be replaced using appropriate imputation methods such as median imputation.

2. Normalisation:

Features such as glucose, BMI, age and blood pressure can have different numerical ranges. Standardisation can transform them to approximately zero mean and unit variance.

3. Encoding:

Categorical variables such as family history can be converted into numerical values, for example:

- No = 0

- Yes = 1

4. Class balancing:

SMOTE is applied only to the training set.

These preprocessing steps improve model consistency, prevent features with larger numerical scales from dominating algorithms, and help the learning algorithm produce more reliable predictions.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels. Justify the root node selection.

A Decision Tree recursively divides the dataset according to feature values. At every node, the algorithm selects the feature that produces the best separation between diabetic and non-diabetic patients.

A suitable tree for this healthcare problem could begin with **Glucose Level**, because glucose is expected to provide strong discrimination between the two classes.

An illustrative first three levels are:

```
                Glucose Level
               /          \
          <= 126           > 126
          /                  \
       BMI                 Glucose Level
      /   \                 /       \
  <= 30    > 30         <= 180      > 180
   /         \             /           \
Non-Diabetic   Risk/Diabetic   Diabetic     Diabetic
```

The exact tree structure and thresholds would depend on the actual dataset.

The root node should be selected using **Information Gain** or **Gini Index**.

For Information Gain:

Information Gain = Entropy(parent) − Weighted Entropy(children)

For Gini Index:

Gini = 1 − Σp_i^2

The best root feature is the one that provides the **highest Information Gain** or **lowest weighted Gini impurity**. If glucose produces the greatest reduction in impurity, it becomes the root node.

(b) Evaluate the model: Accuracy, Precision, Recall and F1-Score. Identify False Negatives and explain how to reduce them.

The important evaluation measures are:

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

F1-Score

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Since the supplied assessment does not contain an actual patient dataset or confusion matrix, exact measured values cannot be calculated from the file. An **illustrative result** could be:

Metric	Example Result
Accuracy	84%
Precision	81%
Recall	88%
F1-Score	84%

A **False Negative (FN)** occurs when the patient is actually diabetic but the model predicts non-diabetic.

False negatives are particularly important in this application because an undiagnosed diabetic patient may not receive timely medical attention.

To reduce false negatives:

- Apply **SMOTE** to improve minority-class learning.
- Use **class weights** to penalise errors on diabetic cases.
- Tune the classification threshold where applicable.
- Optimise for **Recall/Sensitivity** rather than accuracy alone.
- Perform cross-validation to obtain a more reliable model.

Clinically, reducing false negatives is important because early identification allows further testing, monitoring and appropriate medical intervention.

(c) Apply post-pruning and compare pre-pruning vs post-pruning.

A fully grown Decision Tree can become excessively complex and learn noise from the training dataset. This is called **overfitting**.

Pre-pruning limits tree growth during training by setting parameters such as:

- max_depth
- min_samples_split
- min_samples_leaf

Post-pruning first allows the tree to grow and then removes branches that contribute little predictive value.

Cost-complexity pruning uses:

$$R_\alpha(T) = R(T) + \alpha \mid T \mid$$

where:

- $R(T)$= error of the tree
- $\mid T \mid$= number of terminal nodes
- α= complexity penalty

An illustrative comparison:

Model	Training Accuracy	Test Accuracy
Pre-pruned Tree	87%	84%
Post-pruned Tree	89%	86%

These values are illustrative because no dataset is supplied.

For clinical deployment, the **post-pruned tree** may be preferable if validation demonstrates better generalisation and fewer unnecessary branches. A simpler tree is also easier for clinicians to understand and audit.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Logistic Regression and compare it with the Decision Tree.

Logistic Regression is a statistical classification technique suitable for predicting the probability of a binary outcome.

For diabetes prediction:

$$P(Diabetes = 1) = \frac{1}{1 + e^{-z}}$$

where:

$$z = \beta_0 + \beta_1 Age + \beta_2 Glucose + \beta_3 BMI + \cdots$$

The model produces a probability between 0 and 1. A threshold such as 0.5 can then be used to classify the patient.

An illustrative comparison is:

Model	Accuracy	AUC-ROC
Decision Tree	86%	0.87
Logistic Regression	84%	0.89

Again, these are **example values**, not measured results from the supplied file.

A statistical model such as Logistic Regression is preferable when:

- Interpretability is important.

- The relationship between predictors and outcome is reasonably systematic.
- Probability estimates are required.
- We need to understand the direction and strength of individual predictors.
- A simpler model is preferred for clinical decision support.

The **AUC-ROC** is particularly useful because it measures the model's ability to distinguish diabetic from non-diabetic patients across different classification thresholds.

(b) Perform feature importance analysis. Identify the top 3 predictors from both models.

For the Decision Tree, feature importance can be calculated from the reduction in impurity contributed by each feature.

For Logistic Regression, the magnitude and statistical significance of the coefficients can be examined after appropriate preprocessing.

An illustrative result is:

Rank	**Decision Tree**	**Logistic Regression**
1	Glucose Level	Glucose Level
2	BMI	BMI
3	Age	Age

Therefore, both models agree on the top three predictors in this example.

This agreement is reasonable because glucose level is directly related to diabetes diagnosis, while BMI and age provide additional information about metabolic and demographic risk.

However, the actual top-three ranking must be calculated from the supplied patient dataset. The assessment file itself provides only the available feature categories, not their values or trained-model importance scores.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP.

The treatment recommendation problem can be formulated as a **Markov Decision Process (MDP)**.

An MDP consists of **States, Actions, Rewards and Transition Probabilities**.

States

States represent the patient's health condition at a particular time.

For example:

- **S1 – Low Risk**
- **S2 – Moderate Risk**
- **S3 – High Risk**
- **S4 – Poorly Controlled**
- **S5 – Improved/Stable**

The state can be determined using factors such as glucose level, BMI, blood pressure and previous health measurements.

Actions

The available actions are:

- **A1 – Diet**
- **A2 – Exercise**
- **A3 – Medication**
- **A4 – Monitor**

These correspond directly to the treatment actions specified in the assessment.

Reward Function

The reward represents the improvement or deterioration in patient health.

For example:

- Significant improvement = **+10**
- Moderate improvement = **+5**
- No significant change = **0**
- Health deterioration = **−5**
- Serious adverse outcome = **−10**

The reward encourages the agent to select actions that lead to improved health outcomes.

Transition Dynamics

After an action is taken, the patient may move from one health state to another.

For example:

High Risk

|

| Exercise

↓

Moderate Risk

|

| Diet

↓

Improved/Stable

The transition depends on the patient's current condition, selected action and resulting health response.

(b) Apply Q-Learning for at least 5 patient episodes.

Q-Learning learns the expected future reward of taking an action in a particular state.

The update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[r + \gamma \max_{a'} Q(s',a') - Q(s,a) \right]$$

where:

- α= learning rate
- γ= discount factor
- r= immediate reward
- s= current state
- a= selected action
- s'= next state

Assume:

$$\alpha = 0.5,\ \gamma = 0.9$$

Initially all Q-values are 0.

Episode 1 – Example

Suppose:

State = High Risk

Action = Diet

Reward = +5

Next State = Moderate Risk

Assume the maximum Q-value of the next state is 4.

$$Q(H\text{،} ighDiet) = 0 + 0.5[5 + (0.9 \times 4) - 0]$$
$$= 0.5[8.6] = 4.3$$

So:

Q(High Risk, Diet) = 4.3

Episode 2

Suppose:

State = High Risk

Action = Exercise

Reward = +6

Maximum next Q-value = 5

$$Q(H\text{،} ighExercise) = 0 + 0.5[6 + (0.9 \times 5)]$$
$$= 0.5(10.5) = 5.25$$

Therefore:

Q(High Risk, Exercise) = 5.25

Episode 3

Suppose:

State = Moderate Risk

Action = Diet

Reward = +5

Maximum next Q-value = 6

$$Q(M\text{،} oderateDiet) = 0 + 0.5[5 + (0.9 \times 6)]$$
$$= 0.5(10.4) = 5.2$$

Therefore:

Q(Moderate Risk, Diet) = 5.2

Example Q-table

State	**Diet**	**Exercise**	**Medication**	**Monitor**
Low Risk	5.0	6.5	1.0	4.0
Moderate Risk	5.2	7.1	6.8	3.5
High Risk	4.3	5.25	8.2	2.0
Poorly Controlled	2.5	4.0	9.1	1.5

The agent can be trained for **at least five patient episodes**, updating the Q-values after every state-action transition.

The convergence criterion can be:

$$\max_{s,a} \mid Q_{new}(s, a) - Q_{old}(s, a) \mid < \epsilon$$

for a small value such as **ε = 0.01** over several consecutive episodes.

In practice, additional safety constraints would be required before using such a system clinically.

(c) Extract the final learned policy and discuss ethical considerations.

The final policy is obtained by selecting the action with the highest Q-value for each state:

$$\pi(s) = \arg\max_{a} Q(s, a)$$

Using the illustrative Q-table:

Patient State	**Best Action**
Low Risk	Exercise
Moderate Risk	Exercise
High Risk	Medication

Patient State	Best Action
Poorly Controlled	Medication

Thus, the learned policy recommends the action with the highest expected long-term reward for each patient state.

However, **the RL agent should not independently prescribe or change medication in a real clinical environment**. It should function as a decision-support system under appropriate clinical supervision.

Ethical considerations

1. Patient Safety:

Incorrect recommendations could cause serious harm. The system must have safety constraints and human clinical oversight.

2. Bias:

If the training data underrepresents certain patient groups, the learned policy may perform poorly for those groups.

3. Explainability:

Doctors should be able to understand why an action was recommended, especially when treatment decisions are high-risk.

4. Privacy:

Patient information must be securely stored and processed while protecting confidentiality.

5. Human Oversight:

The final treatment decision should remain with qualified healthcare professionals rather than being completely delegated to the RL agent.

6. Continuous Validation:

The model should be regularly evaluated to ensure that its recommendations remain safe and effective.

Overall, reinforcement learning can model personalised treatment decisions, but clinical deployment requires strong validation, safety constraints, fairness checks and human supervision.